

Advances in Computational Biology

What is Bioinformatics?

Mathematical, statistical, and computing approach to solve biological problems using DNA and amino acids sequencing, and related information. Or can be summarized into

1. Body of tools and algorithm to handle **large and complex** biological information
2. New specific discipline created from computer and biology.
3. The field of science where computer science, biology, and information technology, converge into a single discipline.

Bioinformatics Milestone

- In 1951 Pauling and Corey propose the structure of alpha helix.
- In 1953 Watson and Crick propose the double helix model for DNA based on the x-ray data obtained from Franklin and Wilkins.
- In 1955 The sequence of the 1st protein, bovine insulin by F. Sanger.
- In 1965 Margaret Dayhoff starts the Atlas of Protein Sequence and Structure, Dayhoff Atlas.
- In 1972 Protein Sequence Database (PSD) by Margaret Dayhoff.
- In 1977 Stanley Cohen invented DNA Cloning Sanger et al. Invented Cycle Sequencing.
- In 1980 The 1st complete gene sequence for bacteriophage ϕ X174 is published (5,386bp, 9 proteins.).
- In 1983 IBM introduce personal computer and Smith-Waterman algorithm for sequence alignment is published.
- In 1981 The polymerase chain reaction is described by Kary Mullis.
- In 1985 Creation of the GenBank.
- In 1987 Practical Extraction Report is released by Larry Wall, Perl is used to analyze biological assets apart from Python.
- In 1987 The National Center for Biotechnology (NCBI) is established.
- In 1988 The FASTA Algorithm is introduced by Pearson & Lipman.
- In 1990 BLAST program is implemented.
- In 1991 The 1st reference to the word 'Bioinformatics' in science literature.
- In 1998 The Institute for Genomics Center (TIGR) is established by Dr. Craig Venter.
- In 2001 Dr. Craig Venter forms Celera in Rockville and the draft of the human genome is published.
- In 2003 The Completion of the Human Genome Project.

- Identifying all approximately 2k-2.5k genes in human DNA.
- Determine the sequence of the 3 million base pairs.
- Store the information in a database.
- Improve tools for data analysis.
- Transfer related technology to private sector.
- Address ethical, legal, and social issues (ELSA) that might arise from the project.

Bioinformatics and System Biology

3 Aspects of System Biology

1. Genomics : Studies of genetic make-up organism.
2. Proteomics : Studies of the structure and function of proteins. What they do?, What they interact with?, And how they contribute to life processes.
3. Metabolomics : Comprehensive and simultaneous studies of metabolites at the systemic levels and their response to stimuli.

Analysis Involving Nucleic-Acid Sequence

DNA Sequencing provides information about genes and genomes. It does it by using chain-terminating reagents to identify the order of fragments and from this to infer the sequence of bases. For Example :

- Restriction fragment length polymorphisms (RFLPs) identified by Southern Blotting reveal individual differences in DNA.
- DNA Fingerprinting uses probes to locate polymorphic DNA fragments.

DNA Analysis

The polymerase chain reaction accelerates the process of analysis :

1. The polymerase chain reaction (PCR) amplifies a single small DNA fragment using 2 short primers that flank the region to be amplified.
2. Cyclic replication is accomplished via heating and cooling which a key factor is Taq polymerase, which is not denatured at high temperature.

Protein interaction can be detected with 2 hybrid system :

The yeast 2-hybrid system relies on fusion proteins and a reporter gene to study protein-protein interactions.

Genetic Engineering

Expression vectors allow production of specific gene products:

Expression vectors contain the promoters and enhancers necessary to drive expression of

the inserted DNA.

Genes can be introduced across species barriers :

Transgenic organisms can be constructed to express genes is a different species or to create mutations in genes to assess phenotype.

Bioinformatics Application

- Read counting
- Gene discovery
- Small RNA characterization
- SNPs and indels detection
- Comparative transcriptomics
- Fusion genes detection
- Alternative splicing variant detection
- Expression level differences
- System biology and pathway studies
- Species evolution
- Gene mining and discovery

Biological data types and analysis objectives

- Genomics
 - Nucleotide genome sequences, metagenomic sequences.
 - Gene finding, functional annotation, sequence alignment, homology determination, comparative analysis, phylogenetic inferencing, association analysis, mutation functional prediction, species distribution analysis.
- Transcriptomics
 - RNA expression levels, transcription factor binding, chromatin structure information
 - Differential expression, clustering, functional enrichment, transcriptional regulation/causal reasoning
- Proteomics
 - Protein levels, structures, and interaction.
 - Protein identification and functional prediction, structural prediction, structural comparison, molecular dynamic simulation, mutation functional prediction, docking prediction, network analysis.
- Metabolomics
 - metabolite/small molecule levels
 - Pathway/network analysis.
- Imaging
 - Microscopy images, MRI images, CT Scans

- Feature extraction, high content screening
- Cytometry
 - Cell levels and phenotypes
 - Cell population clustering and biomarker discovery
- System Biology
 - All of the above
 - Network analysis, causal reasoning, reverse causal reasoning, drug target prediction, regulatory network analysis, information flow, population dynamics, modeling and simulation.

Big Data in Bioinformatics

New paradigm that transform the studies into large scale research.